



Homework 5



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Pandas Group Operations

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Direction

- Please email your **Python Script** for **Part 1 and Part 2 in Homework 5** with the name below:
 - `danl_210_hw5_LASTNAME_FIRSTNAME.py`
(e.g., `danl_210_hw5_choe_byeonghak.py`)
- The due for **Part 1 and Part 2 in Homework 5** is February 25, Wednesday, 2026, 11:59 P.M.
- The due for **Part 3 in Homework 5** is February 28, Saturday, 2026, 11:59 P.M.
- Please send Prof. Choe an email (bchoe@geneseo.edu) if you have any questions.

Part 1. Spotify Data with Group Operation



Below is `spotify` DataFrame that reads the file `spotify_all.csv` containing data of Spotify users' playlist information (Source: [Spotify Million Playlist Dataset Challenge](#)).

```
spotify = pd.read_csv('https://bcdanl.github.io/data/spotify_all.csv')
```

pid	playlist_name	pos	artist_name
<dbl>	<chr>	<dbl>	<chr>
0	Throwbacks	0	Missy Elliott
0	Throwbacks	1	Britney Spears
0	Throwbacks	2	Beyoncé
0	Throwbacks	3	Justin Timberlake
0	Throwbacks	4	Shaggy
0	Throwbacks	5	Usher
0	Throwbacks	6	Usher
0	Throwbacks	7	The Pussycat Dolls
0	Throwbacks	8	Destiny's Child
0	Throwbacks	9	OutKast
0	Throwbacks	10	Nelly Furtado
0	Throwbacks	11	Jesse McCartney
0	Throwbacks	12	Jesse McCartney
0	Throwbacks	13	Jesse McCartney
0	Throwbacks	14	Cassie
0	Throwbacks	15	Omarion

pid	playlist_name	pos	artist_name	
<dbl>	<chr>	<dbl>	<chr>	
0	Throwbacks	16	Avril Lavigne	
0	Throwbacks	17	Chris Brown	
0	Throwbacks	18	Beyoncé	
0	Throwbacks	19	Destiny's Child	
0	Throwbacks	20	Sheryl Crow	
0	Throwbacks	21	The Black Eyed Peas	
0	Throwbacks	22	Bowling For Soup	
0	Throwbacks	23	The Click Five	
0	Throwbacks	24	Chris Brown	

1-25 of 10,000 rows | 1-4 of 7 columns

Previous 1 2 3 4 5 6 ... 400 Next

Variable Description

- **pid**: playlist ID; unique ID for playlist
- **playlist_name**: a name of playlist
- **pos**: a position of the track within a playlist (starting from 0)
- **artist_name**: name of the track's primary artist
- **track_name**: name of the track
- **duration_ms**: duration of the track in milliseconds
- **album_name**: name of the track's album

Definition of a Song

- In Part 2, a **song** is defined as a combination of a **artist_name** value and a **track_name** value.
- E.g., the following provides the 12 distinct songs with the same **track_name** —I Love You:

artist_name	track_name
<chr>	<chr>
Alex & Sierra	I Love You
Alex Clare	I Love You
Axwell /\ Ingrosso	I Love You
Beat Happening	I Love You
Climax Blues Band	I Love You
Céline Dion	I Love You
Donna Summer	I Love You
Eli Young Band	I Love You
Faith Evans	I Love You
Mary J. Blige	I Love You

artist_name	track_name
<chr>	<chr>
OMFG	I Love You
Said The Whale	I Love You
1-12 of 12 rows	

Question 1

Write code with pandas `groupby()` to calculate the total duration (in minutes) of each playlist and then sort by the total duration in descending order to identify the playlists from longest to shortest total playtime.

Question 2

Write code with pandas `groupby()` to identify the top five songs with the highest frequency of appearances in the `spotify` DataFrame.

Question 3

- Write code with pandas `groupby()` to create a DataFrame that contains information about how often each song occurs within the `spotify` DataFrame.
 - In this new DataFrame, each observation should represent a distinct song.
- Then, identify top 5% songs based on their frequency of appearances.

Question 4

Write code with pandas `groupby()` to list all artists who have more than 50 unique songs in the `spotify` DataFrame.

Question 5

Write code with pandas `groupby()` to create a DataFrame that identifies all the playlists featuring the song “One Dance” by Drake.

Question 6

Write code with pandas `groupby()` to identify the longest and shortest duration songs (based on `duration_ms`) for each unique artist.

Question 7

Write code with pandas `groupby()` to find the same song(s) appearing more than once in the same playlist.

Question 8

Write code with pandas `groupby()` to filter all songs that appear in more than 100 different playlists.

Part 2. Shoes Online Shopping

You searched for “shoes”

30956 items

Sort: Featured

The screenshot displays a grid of eight shoe products. Each product card includes a main image, a row of smaller color swatches, the product name, and the price. The products are:

- adidas Samba OG Sneaker (Women)**: Price range \$100.00 – \$120.00. Marked as 'New!'.
- Nike Dunk Low Basketball Sneaker (Women)**: Price range \$100.00 – \$120.00. Marked as 'New!'. Includes a note: 'Add to Bag or Wish List to see price'.
- On Cloud 6 Sneaker (Women)**: Price \$150.00. Marked as 'New!'.
- Nina Bergen Ankle Strap Sandal (Women)**: Price \$89.00. Marked as 'Sponsored'.
- Munro Lucia Sandal (Women)**: Price range \$99.98 – \$157.46 (Up to 52% off). Original price \$204.95 – \$209.95. Marked as 'New Markdown'.
- adidas Gazelle Bold Platform Sneaker (Women)**: Price range \$100.00 – \$120.00.
- New Balance Gender Inclusive 9060 Sneaker**: Price \$149.99.
- Nina Abbi Slingback Sandal (Women)**: Price \$89.00. Marked as 'Sponsored'.

Below is `shoes` DataFrame that reads the file `onlinestore_shoes_simple.csv` containing data of “shoes” search information from an online store.

```
shoes = pd.read_csv('https://bcdanl.github.io/data/onlinestore_shoes_simple.csv')
```

```
import pandas as pd
import numpy as np
shoes = pd.read_csv('https://bcdanl.github.io/data/onlinestore_shoes_simple.csv')
```

id	brandId	brandName				
<dbl>	<dbl>	<chr>				
2867145	1673	Nike				
3208294	1673	Nike				
3339203	255	Birkenstock				
4361638	1613	Munro				
4680267	1673	Nike				
4908062	12214320	Jewel Badgley Mischka				
5112370	1706	Naturalizer				
5172382	61560	Spring Step				
5272418	1706	Naturalizer				
5476290	12214320	Jewel Badgley Mischka				
5723075	8180	Sam Edelman				
5736814	1656	New Balance				
5883450	52	Aerosoles				
5891793	4053	ASICS®				
5909725	15341	Badgley Mischka Collection				
5918989	81008460	Vionic				
5935983	661	Franco Sarto				
5936789	6129	Vince Camuto				
5988085	8685	LifeStride				
6465879	3046	Linea Paolo				
6593773	342	Burberry				
6753858	8174	Nine West				
6924717	21261	Alexander Wang				
7030184	1673	Nike				
7139562	9423	Tory Burch				

1-25 of 10,000 rows | 1-3 of 8 columns

Previous

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2

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400

Next

Variable Description

Variable	Description
id	Unique identifier for the product
brandId	Unique identifier for the brand
brandName	Name of the brand
name	Product name or title
reviewCount	Number of customer reviews

Variable	Description
<code>reviewStarRating</code>	Average star rating from customer reviews
<code>current_p</code>	Current price
<code>clearance_p</code>	Clearance price (<code>NA</code> if not available for clearance)

- If the product is **on clearance**, the product's `current_p` equals its `clearance_p`.

Question 9

Add a new variable `discount_pct` to the `shoes` DataFrame that shows the percentage markdown for clearance items (0 % for non-clearance).

$$(\text{Discount Percentage}) = 100 \times \frac{(\text{Current Price}) - (\text{Clearance Price})}{(\text{Current Price})}$$

- Note that the value of `discount_pct` is zero for all observations.

Question 10

For each brand, compute the interquartile range (IQR) of its `current_p`.

Question 11

List the top 20 brands with the highest average star rating, considering only brands that have at least 1000 reviews total.

Question 12

For each brand, compute the proportion of products on clearance. Return the result sorted from highest to lowest proportion.

Question 13

For every brand, calculate the correlation between product price (`current_p`) and star rating. Keep only brands with more than 100 products.

Question 14

Identify “premium” products inside each brand as those priced above the 90th percentile of that brand. Among these premium items, what fraction are on clearance overall?

Question 15

After binning `current_p` into price quartiles with `pd.qcut()`, count the number of clearance products for each brand–quartile combination.

- Below explains `pd.qcut()`:

```
shoes["price_qtile"] = pd.qcut(shoes["current_p"], 4, labels=["Q1", "Q2", "Q3", "Q4"])
```

- `pd.qcut(shoes["current_p"], 4, ...)`
 - `qcut` stands for “quantile cut.”
 - Given the series of prices, it computes the 0th, 25th, 50th, 75th, and 100th percentiles.
 - It then assigns each product to one of four bins so that each bin contains (as close as possible to) the same number of products.
- `labels=["Q1", "Q2", "Q3", "Q4"]`
 - Instead of getting the default interval-notation (e.g. (10.0, 25.5]), each bin is labeled “Q1” through “Q4.”
 - “Q1” will be the lowest 25% of prices, “Q2” the next 25%, and so on, up to “Q4” for the top-priced 25%.
- Assignment into `shoes["price_qtile"]`
 - The result is a categorical variable in your `shoes` DataFrame.
 - You can now easily group or filter by “Q1”, “Q2”, etc., rather than working with numeric cut-points.

Question 16

- For each brand in the `shoes` DataFrame, add a new variable `rev_scaled` that re-scales `reviewCount` to the [0, 1] range.

- The product with the highest `reviewCount` within its brand has a value 1 in the `rev_scaled` variable.
- The product with the lowest `reviewCount` within its brand has a value 0 in the `rev_scaled` variable.

Part 3: Enhance Your Blogs with Data Visualizations

Tip: Consider using generative AI to polish your visualizations.

Spotify Blog (Homework 2)

Add the following section to your **Spotify** blog:

- **Code**
 1. Use pandas to identify the ten artists with the most tracks.
 2. Use seaborn to plot the distribution of `pos` for each of those artists.
- **Description**

Write 2–3 sentences explaining how the `pos` distribution differs across these top ten artists.

Holiday Movie Blog (Homework 3)

Add the following section to your **Holiday Movie** blog:

- **Code**
 1. Use pandas to select the five genres with the highest film counts.
 2. Use numpy to add a new variable, log of `num_votes`.
 3. Use seaborn to plot the relationship between log of `num_votes` and `average_rating`, grouping by genre and coloring or faceting by `title_type`.
- **Description**

Write 2–3 sentences describing how the log of `num_votes` vs. `average_rating` relationship varies among the top five genres and between different `title_type`s.